

PHYSICS

REST AND MOTION

Complete JEE Main Literature

Kinematics | Equations of Motion | Graphs | Projectile Motion | Circular Motion

Covers: NCERT + JEE Main PYQ concepts + All Formulae + Diagrams

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§1 Basic Concepts: Rest, Motion & Frame of Reference

1.1 What is Rest?

A body is said to be **at rest** if it does not change its position with respect to its surroundings (reference point) over time. Rest is a *relative* concept — a body at rest in one frame of reference may be in motion in another.

1.2 What is Motion?

A body is said to be **in motion** if it changes its position with respect to its surroundings over time. Like rest, motion is also a relative concept.

Rest & Motion are RELATIVE — depend on frame of reference

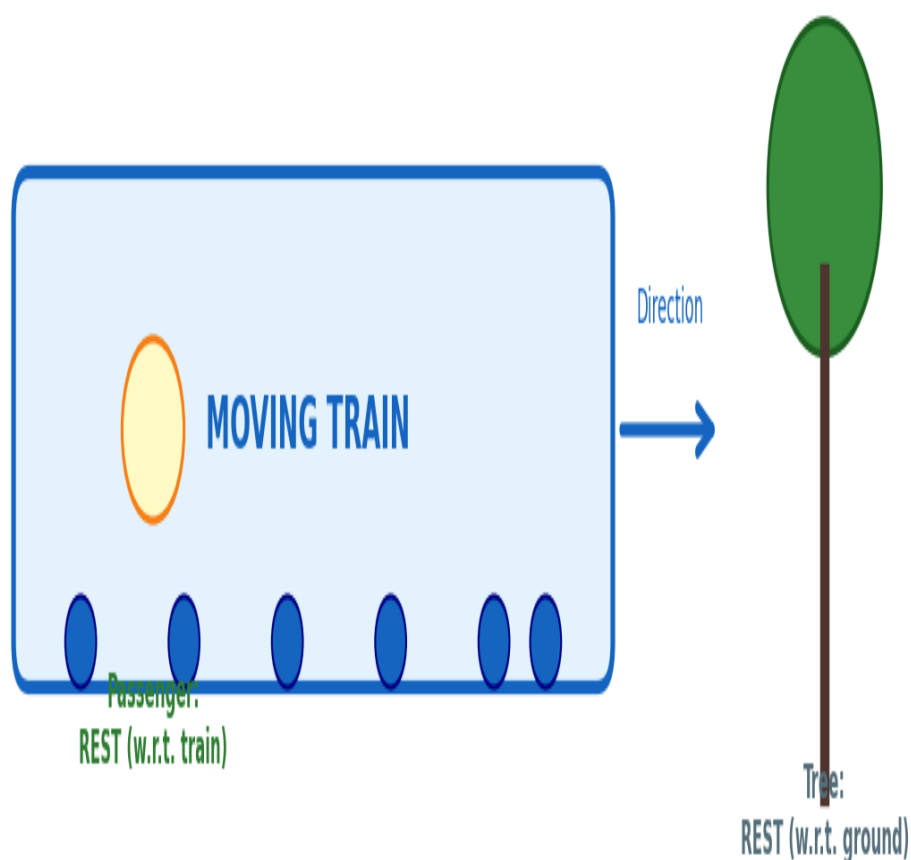


Fig 1.1 — A passenger in a moving train is at rest w.r.t. the train but in motion w.r.t. the ground.

1.3 Frame of Reference

A **frame of reference** is a coordinate system with a clock, used to describe the position and motion of objects. It consists of three mutually perpendicular axes (X, Y, Z) and an origin.

Types of Frames of Reference

- **Inertial Frame:** A frame at rest or moving with constant velocity (no acceleration). Newton's laws hold here.
- **Non-Inertial Frame:** An accelerating frame. Pseudo forces must be introduced.
 - *For JEE Main, the ground is usually taken as the inertial frame of reference.*

1.4 Types of Motion

Type	Description	Example
Rectilinear (1D)	Along a straight line	Falling stone, car on highway
Circular	Along a circular path	Earth around Sun, fan blade
Oscillatory/Periodic	To-and-fro about mean position	Pendulum, spring-mass
Random	No fixed path or direction	Molecules in gas, Brownian motion
Translatory	Every particle has identical motion	Sliding block
Rotatory	Body rotates about a fixed axis	Spinning top, Earth's spin

★ JEE
Tip
Rest and motion are relative. In JEE, always identify the frame of reference first.

§2 Distance & Displacement

2.1 Distance

Distance is the total length of the path actually travelled by a body, regardless of direction. It is a **scalar** quantity and is always **positive or zero**. SI unit: **metre (m)**.

Key properties of Distance:

- Always ≥ 0 ; never negative.
- Depends on the path taken.
- Can only increase with time.
- Measured by odometer in vehicles.

2.2 Displacement

Displacement is the shortest straight-line distance from the initial position to the final position, with a specified direction. It is a **vector** quantity. SI unit: **metre (m)**.

Key properties of Displacement:

- Can be positive, negative, or zero.
- Depends only on initial and final positions.
- Independent of path taken.
- Displacement = 0 if body returns to start.
- $|\text{Displacement}| \leq \text{Distance}$ (always).

Distance vs Displacement

DISTANCE (Scalar)

DISPLACEMENT (Vector)

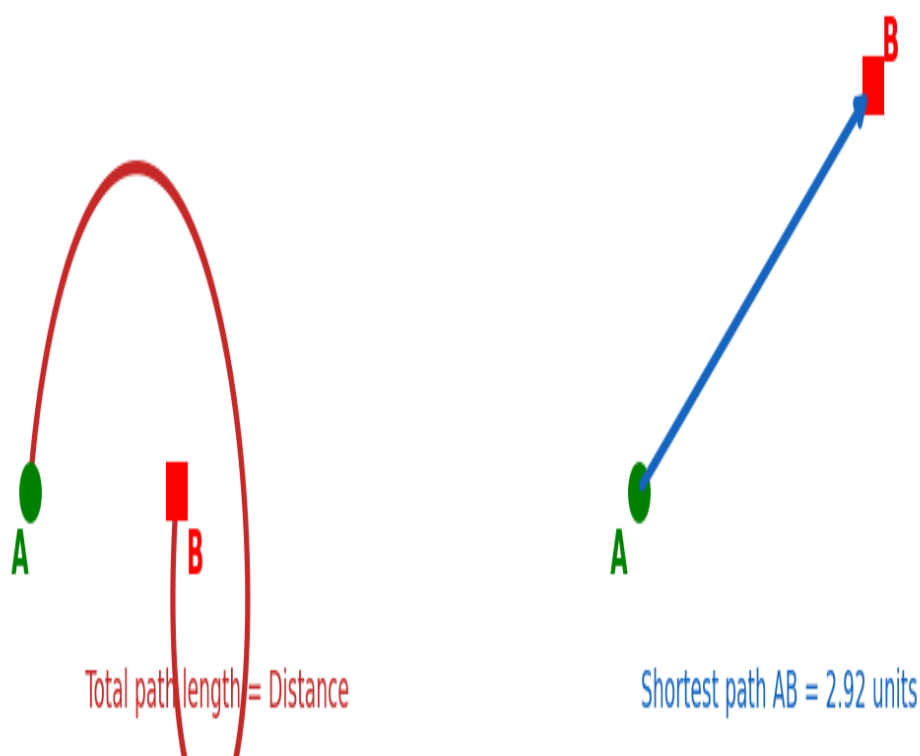


Fig 2.1 — Distance (red winding path) vs Displacement (blue straight arrow from A to B).

Displacement = Final position – Initial position

If $A = (x_1, y_1)$ and $B = (x_2, y_2)$: $|\Delta r| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

2.3 Worked Examples

Example	Distance	Displacement
Walk 4m East then 3m North	7 m	5 m (NE, 37° from East)
Complete circular loop (radius r)	$2\pi r$	0
Semicircle of radius r	πr	$2r$ (diameter)
Oscillation: $A \rightarrow B \rightarrow A$ (amplitude A)	$4A$ (2 cycles)	0

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Tip

JEE often asks: "displacement = 0 but distance $\neq 0$ " — circular/oscillatory motion.

Also: minimum distance = |displacement|.

§3 Speed & Velocity

3.1 Speed

Speed is the rate of change of distance. It is a **scalar** quantity. SI unit: **m/s**. It is always ≥ 0 .

$$\text{Speed} = \text{Distance} / \text{Time}$$

$$\text{Average Speed} = \text{Total Distance} / \text{Total Time}$$

Types of Speed:

Type	Definition	Formula
Uniform Speed	Equal distances in equal time intervals	$v = \text{const}$
Non-Uniform Speed	Unequal distances in equal time intervals	$v \neq \text{const}$
Average Speed	Total distance / total time	$v_{\text{avg}} = d_{\text{total}} / t_{\text{total}}$
Instantaneous Speed	Speed at a specific instant	$v_{\text{inst}} = \lim(\Delta t \rightarrow 0) \Delta d / \Delta t$

3.2 Velocity

Velocity is the rate of change of displacement. It is a **vector** quantity. SI unit: **m/s**. It can be positive, negative, or zero.

$$\text{Velocity} = \text{Displacement} / \text{Time}$$

$$\mathbf{v} = \Delta \mathbf{r} / \Delta t$$

$$\text{Average Velocity} = \text{Total Displacement} / \text{Total Time}$$

3.3 Key Differences: Speed vs Velocity

Property	Speed	Velocity
Type	Scalar	Vector
Formula	Distance/Time	Displacement/Time
Can be zero?	Only if at rest	Yes (even in motion)
Negative?	Never	Yes
Uniform circular motion	Constant speed	Continuously changing velocity
Relationship	$\text{speed} \geq \text{velocity} $	$ \text{velocity} \leq \text{speed}$

3.4 Unit Conversion

1 km/h = 5/18 m/s ■ To convert km/h $\rightarrow$ m/s, multiply by 5/18.

1 m/s = 18/5 km/h = 3.6 km/h ■ To convert m/s $\rightarrow$ km/h, multiply by 3.6.

★ JEE

Tip

*A body
in uniform
circular
motion
has constant
SPEED but
changing
VELOCITY*

*—
hence
it has an
acceleration!
This is
tested
in JEE
almost
every
year.*

§4 Acceleration & Retardation

4.1 Definition

Acceleration is the rate of change of velocity with respect to time. It is a **vector** quantity. SI unit: **m/s²**.

$$a = \Delta v / \Delta t = (v - u) / t$$

$$\text{Instantaneous acceleration: } a = dv/dt = d^2x/dt^2$$

4.2 Types of Acceleration

Type	Description
Uniform Acceleration	Velocity changes by equal amounts in equal time intervals. $a = \text{constant}$.
Non-Uniform Acceleration	Rate of change of velocity is not constant.
Zero Acceleration	Body moves with constant velocity (or is at rest).
Retardation / Deceleration	Acceleration opposite to velocity; body slows down. a is negative.
Centripetal Acceleration	In circular motion, directed towards centre. $a_c = v^2/r = \omega^2 r$.

Important Note on Signs

- If v and a are in the same direction $\rightarrow$ body speeds up.
- If v and a are in opposite directions $\rightarrow$ body slows down (retardation).
- Zero velocity does NOT mean zero acceleration. (E.g., ball at highest point in vertical throw: $v=0$ but $a=g$ downward)

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Tip

A body can have zero velocity but non-zero acceleration

—
classic JEE trap. At the highest point of a vertical throw, $v=0$ but $a=9.8 \text{ m/s}^2$ downwards.

§5 Equations of Motion (Uniformly Accelerated)

These three equations (+ one for displacement in n^{th} second) apply when acceleration is **constant**. u = initial velocity, v = final velocity, a = acceleration, t = time, s = displacement.

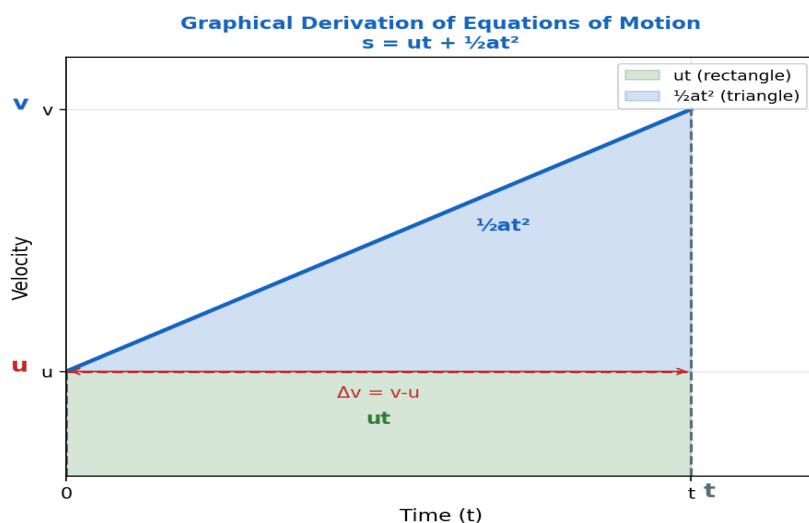


Fig 5.1 — Graphical derivation of $s = ut + \frac{1}{2}at^2$ from velocity-time graph.

5.1 The Three (+ One) Equations

1st: $v = u + at$

2nd: $s = ut + \frac{1}{2}at^2$

3rd: $v^2 = u^2 + 2as$

4th (nth second): $S_n = u + a(2n - 1)/2$

Kinematic Equations of Uniformly Accelerated Motion

Equation	Variables Involved	Missing
$v = u + at$	v, u, a, t	s
$s = ut + \frac{1}{2}at^2$	s, u, a, t	v
$v^2 = u^2 + 2as$	v, u, a, s	t
$s = (u+v)t/2$	s, u, v, t	a
$S_n = u + a(2n-1)/2$	S_n, u, a, n	—

Fig 5.2 — Summary of kinematic equations and the variable each eliminates.

5.2 Derivations

1st Equation: $v = u + at$

Acceleration $a = (v-u)/t$ ■ $at = v-u$ ■ $v = u + at$. This is directly from the definition of acceleration.

2nd Equation: $s = ut + \frac{1}{2}at^2$

Average velocity $= (u+v)/2 = (u + u+at)/2 = u + \frac{1}{2}at$. Displacement $s = \text{average velocity} \times t = (u + \frac{1}{2}at)t = ut + \frac{1}{2}at^2$.

3rd Equation: $v^2 = u^2 + 2as$

From 1st: $t = (v-u)/a$. Substitute in 2nd: $s = u(v-u)/a + \frac{1}{2}a(v-u)^2/a^2 = (v^2-u^2)/(2a)$. Hence $v^2 = u^2 + 2as$.

4th Equation: $S_n = u + a(2n-1)/2$

Distance in n seconds: $S_n = un + \frac{1}{2}an^2$. Distance in $(n-1)$ seconds: $S_{(n-1)} = u(n-1) + \frac{1}{2}a(n-1)^2$.

Displacement in n th second = $S_n - S_{(n-1)} = u + a(2n-1)/2$.

Displacement in n th second: $S_n = u + a(2n-1)/2$
 $(u=0, a=2 \text{ m/s}^2) - \text{ratio } 1:3:5:7:9\ldots$

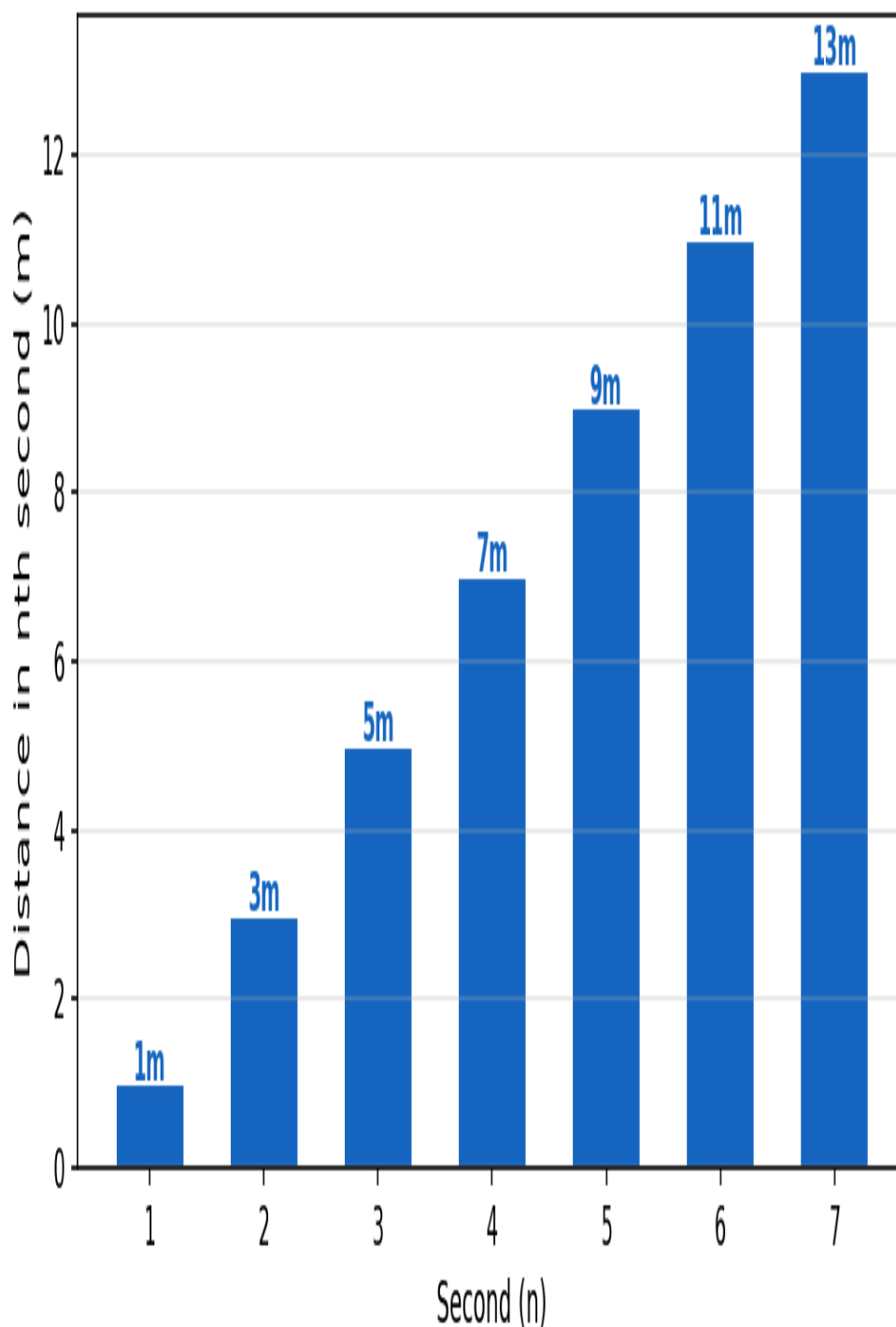


Fig 5.3 — Displacement in consecutive seconds from rest: ratio 1:3:5:7:...

5.3 Important Results

- For a body starting from rest ($u=0$): $v = at$, $s = \frac{1}{2}at^2$, $v^2 = 2as$.
- For a body coming to rest ($v=0$): $0 = u + at \rightarrow t = u/a$; $0 = u^2 + 2as \rightarrow s = u^2/2a$.
- Ratio of distances in 1st, 2nd, 3rd... seconds (from rest): 1:3:5:7:... (odd number ratio).
- Ratio of distances in 1st, 2nd, 3rd... time intervals (from rest): 1:4:9:... (perfect squares)

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Tip

The "odd number rule" (1:3:5:7...) for displacement in successive seconds from rest is a direct JEE application. Derive from $S \propto t^2$ for $u=0$.

§6 Motion Under Gravity

Near Earth's surface, all objects experience a constant downward acceleration due to gravity: $g = 9.8 \text{ m/s}^2 \approx 10 \text{ m/s}^2$ (for JEE calculations). Air resistance is usually neglected.

6.1 Free Fall (Body Released from Rest)

Take downward as positive, $u = 0$:

$$v = gt$$

$$h = \frac{1}{2}gt^2$$

$$v^2 = 2gh$$

$$\text{Time to fall height } h: t = \sqrt{(2h/g)}$$

Free Fall under Gravity ($g = 9.8 \text{ m/s}^2$, $u = 0$)

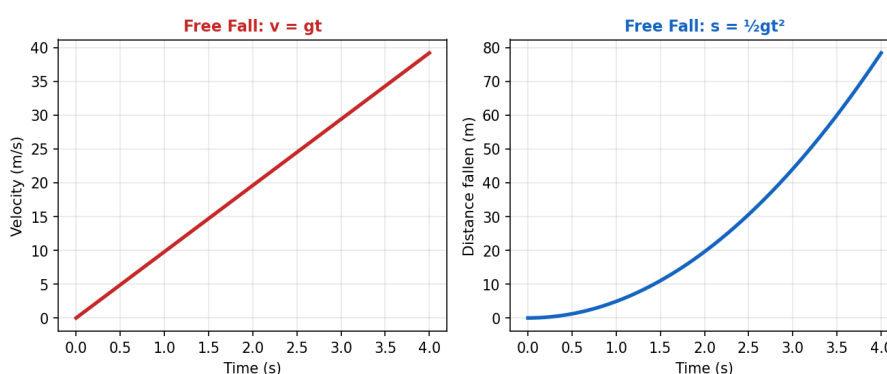


Fig 6.1 — Velocity ($v=gt$) and distance fallen ($h=\frac{1}{2}gt^2$) vs time in free fall.

6.2 Body Thrown Vertically Upward

Initial velocity = u (upward). Take upward as positive, then $a = -g$.

$$\text{Time to reach maximum height: } t_{\text{up}} = u/g$$

$$\text{Maximum height: } H = u^2/2g$$

$$\text{Total time of flight: } T = 2u/g$$

$$\text{Velocity on return to same level: } v = -u \text{ (equal magnitude, opposite direction)}$$

Key Symmetry of Vertical Throw

- Time of ascent = Time of descent = u/g .
- Speed at a given height is same during ascent and descent.
- At highest point: $v = 0$, but $a = g$ (downward) — velocity is zero, acceleration is not!
- Body thrown up with u returns to same level with speed u .

6.3 Body Thrown Down from Height h

$$v^2 = u^2 + 2gh \text{ (} u = \text{initial downward speed)}$$

$$h = ut + \frac{1}{2}gt^2$$

$$v = u + gt$$

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Tip

In free
fall:
ratio of
distanc
es in
1st,
2nd,
3rd sec
=

1:3:5.

The
time to
fall
from
height

$H = t$;
to fall
from
 $4H =$
 $2t$.

Time is
proport
ional to
 $\sqrt{h}$.

§7 Graphical Analysis of Motion

7.1 Distance–Time (d–t) Graph

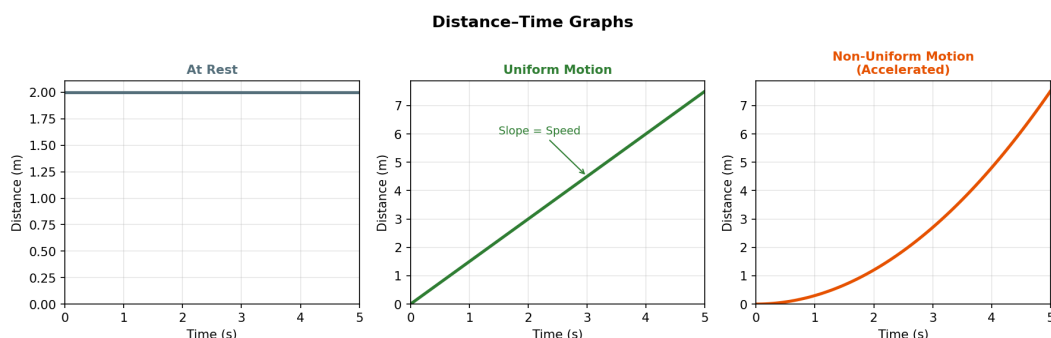


Fig 7.1 — d–t graphs: at rest (horizontal), uniform motion (straight line), accelerated (curve).

Shape of d–t Graph	Type of Motion	Slope = Speed
Horizontal line (slope = 0)	At Rest	0
Straight line (positive slope)	Uniform motion	Constant v
Concave up curve (slope $\uparrow$)	Uniformly accelerated	Increasing
Concave down curve (slope $\downarrow$)	Uniformly decelerated	Decreasing
Negative slope (not physical)	Not possible (distance can't decrease)	—

7.2 Velocity–Time (v–t) Graph

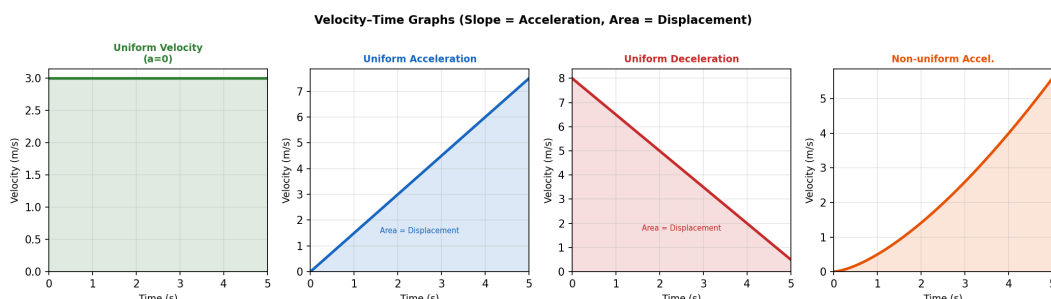


Fig 7.2 — v–t graphs: uniform velocity, uniform acceleration, deceleration, non-uniform acceleration.

Slope of v–t graph = Acceleration ($a = \Delta v / \Delta t$)

Area under v–t graph = Displacement

Shape of v–t Graph	Slope (= a)	Area (= s)
Horizontal line	0 (uniform velocity)	Rectangle = vt
Straight line (+ slope)	Constant +a	Triangle/trapezium
Straight line (– slope)	Constant –a (deceleration)	Trapezium
Curve (slope $\uparrow$)	Increasing a	Curved region
Straight line through origin	Const a (body starts from rest)	$\frac{1}{2}bt \cdot vt$

★ JEE

Tip

In JEE, $v-t$ graphs are tested heavily. Remember: area = displacement (can be negative if curve goes below x -axis), slope = acceleration.

§8 Relative Motion

The velocity of object A with respect to object B is called **relative velocity**.

$$\underline{v}_{AB} = \underline{v}_A - \underline{v}_B \text{ (velocity of A relative to B)}$$

$$\underline{v}_{BA} = \underline{v}_B - \underline{v}_A \text{ (velocity of B relative to A)}$$

$$\underline{v}_{AB} = -\underline{v}_{BA}$$

Relative Velocity

Same Direction: $\underline{v}_{rel} = \underline{v}_A - \underline{v}_B$

Opposite Direction: $\underline{v}_{rel} = \underline{v}_A + \underline{v}_B$

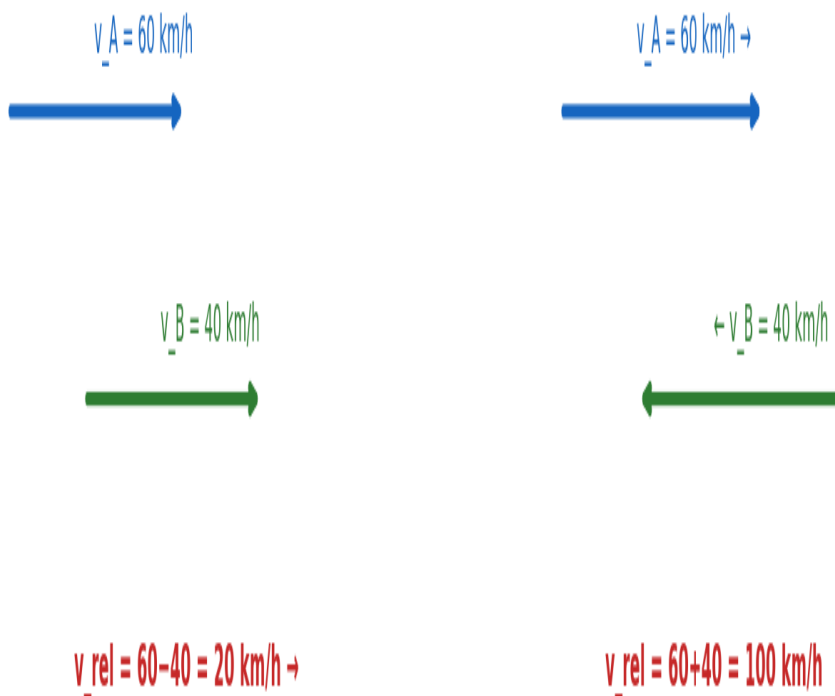


Fig 8.1 — Relative velocity in same direction (subtract) and opposite direction (add).

8.1 Cases

- **Same direction ($\underline{v}_A > \underline{v}_B$):** $\underline{v}_{rel} = \underline{v}_A - \underline{v}_B \rightarrow$ Slow approach

- **Opposite directions:** $v_{rel} = v_A + v_B \rightarrow$ Fast approach
- **A at rest:** $v_{rel} = -v_B \rightarrow$ B appears to move backward
- **Equal velocity, same direction:** $v_{rel} = 0 \rightarrow$ Appear at rest to each other

8.2 River-Boat Problem (2D Relative Motion)

A boat of speed v_b (in still water) crosses a river of width d where water flows at v_r . To cross in minimum time: aim perpendicular to river banks.

Minimum time to cross: $t_{min} = d / v_b$

Drift (displacement downstream) = $v_r \times t_{min} = v_r \cdot d / v_b$

To cross in minimum distance: aim at angle θ upstream where $\sin\theta = v_r / v_b$

Minimum distance: d (perpendicular) only if $v_b \geq v_r$

★ JEE

Tip

River-boat is a staple JEE problem. Two classic cases: (1) Minimum TIME —boat perpendicular, drift = $v_r \cdot d / v_b$. (2) Minimum DISTANCE —boat aimed upstream at $\sin\theta = v_r / v_b$.

§9 Projectile Motion

A **projectile** is any object given an initial velocity and then left to move under gravity alone (no air resistance). The motion is a combination of uniform horizontal motion and uniformly accelerated vertical motion.

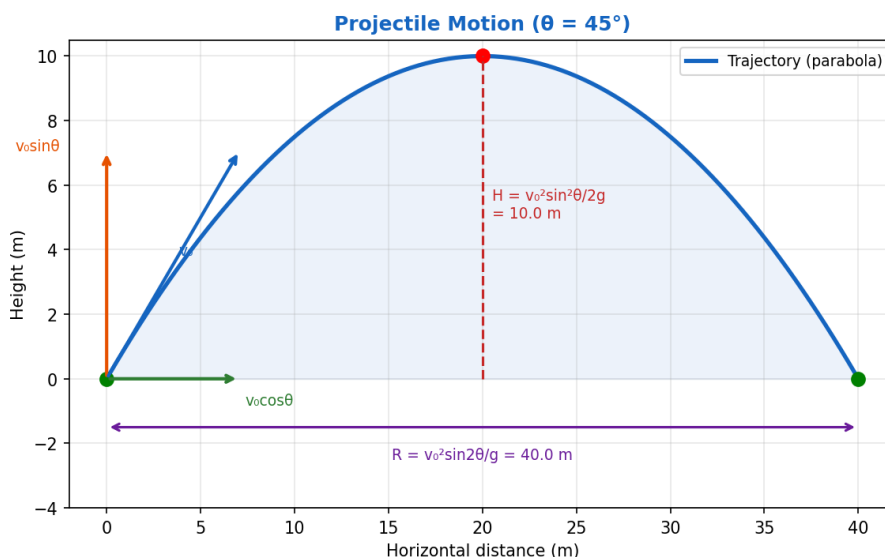


Fig 9.1 — Projectile trajectory (parabola), showing components, maximum height H , and range R .

9.1 Equations of Projectile Motion

For initial speed u , angle θ with horizontal, $g = 9.8 \text{ m/s}^2$:

Horizontal: $x = u \cdot \cos \theta \cdot t$ (uniform, no acceleration)

Vertical: $y = u \cdot \sin \theta \cdot t - \frac{1}{2}gt^2$ (accelerated, $a = -g$)

Velocity components: $v_x = u \cdot \cos \theta$, $v_y = u \cdot \sin \theta - gt$

Speed at time t : $v = \sqrt{(v_x)^2 + (v_y)^2}$

9.2 Key Results

Time of flight: $T = 2u \cdot \sin \theta / g$

Maximum height: $H = u^2 \cdot \sin^2 \theta / 2g$

Horizontal range: $R = u^2 \cdot \sin 2\theta / g$

Maximum range (at $\theta=45^\circ$): $R_{\text{max}} = u^2/g$

Equation of trajectory: $y = x \cdot \tan \theta - gx^2 / (2u^2 \cos^2 \theta)$ [Parabola]

9.3 Important Properties

- Horizontal velocity ($v_x = u \cdot \cos \theta$) remains constant throughout (no air resistance).
- Vertical velocity decreases going up and increases going down.
- At maximum height: $v_y = 0$ (but v_x is unchanged).
- Trajectory is a parabola.
- For same speed, angles θ and $(90^\circ - \theta)$ give the same range.
- Range is maximum at $\theta = 45^\circ$; H is max at $\theta = 90^\circ$.

- Horizontal range $R = 4H \cdot \cot\theta = 4H \cdot (\cos\theta/\sin\theta)$.
- For $\theta=45^\circ$: $R = 4H$ (range = 4 times max height).

9.4 Horizontal Projectile (Thrown Horizontally from Height h)

Time to hit ground: $t = \sqrt{(2h/g)}$

Horizontal distance (range): $R = u \cdot \sqrt{(2h/g)}$

Velocity at ground: $v = \sqrt{(u^2 + 2gh)}$

Angle with horizontal: $\tan\alpha = v_y/v_x = gt/u$

★ JEE

Tip

JEE frequently tests: R same for θ and $(90^\circ - \theta)$; $\theta=45^\circ$ for R_{max} ; "at what angle should you throw for max range on an inclined plane?" — inclined projectile.

§10 Uniform Circular Motion (UCM)

In **Uniform Circular Motion**, a body moves along a circular path with constant **speed**. Although speed is constant, velocity continuously changes direction — hence there is acceleration.

Uniform Circular Motion
Speed constant, velocity direction changes

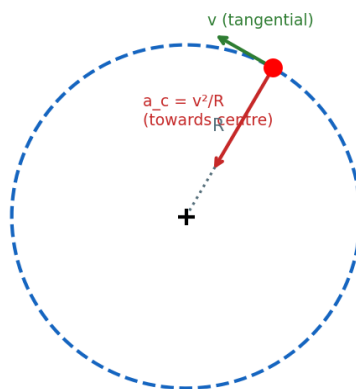


Fig 10.1 — UCM: velocity is tangential, centripetal acceleration points to centre.

10.1 Key Quantities

Angular velocity: $\omega = 2\pi/T = 2\pi f$ (rad/s)

Linear speed: $v = r\omega$

Centripetal acceleration: $a_c = v^2/r = \omega^2 r$ (towards centre)

Centripetal force: $F_c = mv^2/r = m\omega^2 r$ (towards centre)

Time period: $T = 2\pi r/v = 2\pi/\omega$

Frequency: $f = 1/T$

Quantity	Symbol	Formula	SI Unit
Angular displacement	θ	arc length / radius = s/r	radian (rad)
Angular velocity	ω	$2\pi/T = 2\pi f = d\theta/dt$	rad/s
Linear speed	v	$r\omega = 2\pi r/T$	m/s
Centripetal acc.	a_c	$v^2/r = \omega^2 r$	m/s^2
Centripetal force	F_c	$mv^2/r = m\omega^2 r$	N

Important Facts about UCM

- Velocity and centripetal acceleration are always perpendicular.
- Centripetal force does NO work (force $\perp$ displacement at every instant).
- No change in kinetic energy in UCM (speed constant).

- The body has acceleration (centripetal) even though speed is constant.
- Example: Earth around Sun, electron around nucleus (classical), car rounding a bend.

★ JEE

Tip

*UCM:
 v and
 a_c
are per
pendic
ular. C
entripe
tal
force is
NOT a
new
force
— it is
provid
ed by
gravity
(satellit
es),
tension
(stone
in
circle),
friction
(car on
road),
etc.*

§11 All Important Formulae at a Glance

Kinematics (uniform a)

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$S_n = u + a(2n-1)/2$$

$$s = (u+v)t/2$$

Free Fall / Vertical Throw

$$v = u \pm gt \text{ (+ downward)}$$

$$h = ut \pm \frac{1}{2}gt^2$$

$$v^2 = u^2 \pm 2gh$$

$$H_{\text{max}} = u^2/2g \text{ (throw up)}$$

$$T_{\text{flight}} = 2u/g$$

$$t_{\text{fall}} = \sqrt{(2h/g)}$$

Projectile Motion

$$T = 2u \cdot \sin\theta/g$$

$$H = u^2 \sin^2\theta/2g$$

$$R = u^2 \sin 2\theta/g$$

$$R_{\text{max}} = u^2/g \text{ (at } \theta=45^\circ)$$

$$y = x \cdot \tan\theta - gx^2/2u^2 \cos^2\theta$$

Uniform Circular Motion

$$v = r\omega$$

$$\omega = 2\pi/T = 2\pi f$$

$$a_c = v^2/r = \omega^2 r$$

$$F_c = mv^2/r = m\omega^2 r$$

$$T = 2\pi r/v$$

Relative Motion

$$v_{AB} = v_A - v_B$$

$$\text{Same direction: } v_{\text{rel}} = v_A - v_B$$

$$\text{Opposite: } v_{\text{rel}} = v_A + v_B$$

$$\text{River min-time drift} = v_r \cdot d/v_b$$

Unit Conversions

$$1 \text{ km/h} = 5/18 \text{ m/s}$$

$$1 \text{ m/s} = 3.6 \text{ km/h}$$

$$g = 9.8 \text{ m/s}^2 \approx 10 \text{ m/s}^2 \text{ (JEE)}$$

§12 JEE Main — Key Concepts & Previous Year Patterns

Graphs

- Slope of $x-t$ graph = velocity; slope of $v-t$ graph = acceleration.
- Area under $v-t$ graph = displacement (can be negative if v is negative).
- A curved $x-t$ graph (concave up) $\rightarrow$ uniformly accelerated motion.
- If $x-t$ graph is a straight line with negative slope $\rightarrow$ moving in $-ve$ direction with constant speed.
- If $v-t$ graph is below x -axis $\rightarrow$ body moving in opposite direction.

Equations of Motion

- **S■** formula: displacement in n th SECOND is $u + a(2n-1)/2$.
- From rest, distances in successive seconds are in ratio 1:3:5:7...
- Total distance in n seconds from rest: proportional to n^2 (1:4:9:16...).
- A body thrown up returns to same level in same time with same speed.
- Braking distance $\propto u^2$ (hence doubling speed quadruples braking distance).

Projectile

- Complementary angles (θ and $90^\circ - \theta$) have the same range.
- At $\theta = 45^\circ$: $R = 4H$ (range = 4 times max height). $H = R/4$.
- Velocity is horizontal at peak; horizontal component never changes.
- Time of flight $T = 2u \sin \theta / g$; depends only on vertical component.
- For horizontal throw from height h : $t = \sqrt{2h/g}$, $R = u\sqrt{2h/g}$.

Circular Motion

- Speed constant in UCM but velocity (vector) changes every instant.
- Centripetal force always points inward and does zero work.
- At top of vertical circle: $mg + N = mv^2/r$ (minimum speed: $v = \sqrt{gr}$).
- Centripetal acceleration $= v^2/r = \omega^2 r$.
- Angular displacement in one revolution $= 2\pi$ rad.

Common Traps / Mistakes

- Distance can NEVER decrease; displacement CAN decrease.
- $v=0$ does NOT imply $a=0$ (e.g., peak of vertical throw).
- Average speed $\neq$ average velocity in general.
- For uniform circular motion: speed constant but v not constant $\rightarrow a \neq 0$.
- **S■** gives displacement in n th SECOND, not total displacement in n seconds.

Quick Memory Aid — Kinematic Equations

velocity eq (no **s**): $v = u + at$

space eq (no **v**): $s = ut + \frac{1}{2}at^2$

velocity-space (no **t**): $v^2 = u^2 + 2as$

n th second (no total **s** or **v**): **S■** $= u + a(2n-1)/2$

This document covers the complete JEE Main syllabus for the chapter Rest and Motion (Kinematics). Practice problems from H.C. Verma (Part 1), D.C. Pandey (Mechanics Part 1), and JEE Main PYQs for mastery of this chapter.

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